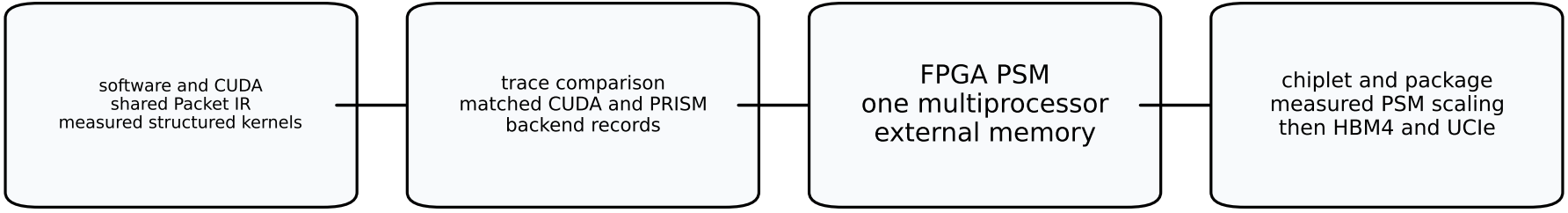
Implementation path



Each stage preserves the same descriptor format, operator contracts, and certificate semantics.